Su Li

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ACADEMIC APPOINTMENTS

Rice University / The University of Texas MD Anderson Cancer Center, Houston, TX
Joint appointment
Postdoctoral Fellow, Rice University *Aug 2025–Present*
Department of Computational Applied Mathematics and Operations Research
Research Collaborator, MD Anderson Cancer Center *Aug 2025–Present*
Advisors: Dr. Andrew J. Schaefer (Rice) and Dr. Iakovos Toumazis (MD Anderson)

University of Toronto, Toronto, Canada *Jul 2024–Jul 2025*
Postdoctoral Fellow, Department of Mechanical and Industrial Engineering
Supervisors: Dr. Vahid Sarhangian, Dr. Adam Diamant, and Dr. André Augusto Ciré

EDUCATION

Texas A&M University, College Station, TX *Sep 2019–May 2024*
Ph.D. in Industrial Engineering GPA: 4.00/4.00
Advisor: Dr. Hrayer Aprahamian

Georgia Institute of Technology, Atlanta, GA *Aug 2017–Dec 2018*
M.S. in Industrial Engineering GPA: 3.60/4.00

The Hong Kong Polytechnic University, Hong Kong SAR *Sep 2013–Jun 2017*
B.Eng. in Industrial and Systems Engineering GPA: 3.76/4.00; First Class Honors
Minor in Software Engineering

RESEARCH AREAS

Optimization under uncertainty; combinatorial and network optimization; Markov decision processes and approximate dynamic programming; healthcare operations and public-health decision-making.

GRANT AND PROPOSAL EXPERIENCE

Collaborated on the development of two National Institutes of Health (NIH) proposals and one National Science Foundation (NSF) proposal.

PEER-REVIEWED JOURNAL ARTICLES

1. **Li, S.**, Aprahamian, H., & Chatterjee, S. (2026). A convex relaxation-based spatial branching approach for optimal robust group testing designs under prevalence rate and dilution behavior uncertainty. *INFORMS Journal on Computing*. Article in Advance.
2. **Li, S.**, & Aprahamian, H. (2024). An optimization-based framework to minimize the spread of diseases in social networks with heterogeneous nodes. *IISE Transactions*, 56(2), 128–142.
3. **Li, S.**, & Aprahamian, H. (2024). Quantifying the benefits of customized vaccination strategies: A network-based optimization approach. *Naval Research Logistics*, 71(1), 64–86.
4. **Li, S.**, Aprahamian, H., Nouiehed, M., & El-Amine, H. (2024). An optimization-based order-and-cut approach for fair clustering of data sets. *INFORMS Journal on Data Science*, 3(2), 124–144.
5. Barth, J., **Li, S.**, Aprahamian, H., & Gupta, D. (2024). Spatiotemporal vaccine allocation policies for epidemics with behavioral feedback dynamics. *Naval Research Logistics*, 71(1), 109–139.

SUBMITTED/WORKING PAPERS

1. **Li, S.**, Ciré, A., Diamant, A., & Sarhangian, V. (2026). Revealing state-relevance weights in dual approximate linear programming. *Submitted to Operations Research.*
2. **Li, S.**, Schaefer, A. J., Jaffray, D. A., Siewerdsen, J. H., & Yung, J. P. (2026). Digital twin and optimization-guided MRI patient sequencing to reduce changeover and increase appointment-slot availability. *Submitted to JCO Oncology Practice.*
   - Selected for Best of Professional at AAPM 2026 (American Association of Physicists in Medicine)
   - Accepted at RSNA 2026 (Radiological Society of North America) Annual Meeting
3. **Li, S.**, Toumazis, I., Schaefer, A., Ishizawa, S., Ozik, J., Collier, N. T., & Meza, R. (2026). A decision-analytic model to optimize diagnostic management for screen-detected pulmonary nodules.
   - Accepted at WCLC 2026 (World Conference on Lung Cancer)
4. **Li, S.**, Jang, I., Barbon, C. E. A., Moreno, A. C., Schaefer, A. J., Lee, J. J., Lai, S. Y., Fuller, C. D., Hutcheson, K. A., & Manduchi, B. (2026). Tumor burden and early swallow recovery predict imaging-graded dysphagia at 2- and 5-years after treatment for oropharyngeal cancer. *Submitted to Radiotherapy & Oncology.*
5. **Li, S.**, Powers, S., & Schaefer, A. J. (2026). Optimizing automated ball-strike challenges in Major League Baseball: A win-probability Markov decision model. *Working paper.*
6. **Li, S.**, & Schaefer, A. J. (2026). Shape-constrained approximate linear programming for integer-resource Markov decision processes via Chvátal–Gomory basis functions. *Working paper.*
7. **Li, S.**, Toumazis, I., & Schaefer, A. J. (2026). State-action-induced nested Markov decision processes: Boundary-price decomposition and policy-column algorithms. *Working paper.*

INVITED CONFERENCE PRESENTATIONS

- INFORMS Annual Meeting, Atlanta, GA, 2025. Invited presentation.
- CORS Annual Conference, Edmonton, Canada, 2025. Invited presentation.
- INFORMS Computing Society Conference, Toronto, Canada, 2025. Invited presentation.
- INFORMS Annual Meeting, Seattle, WA, 2024. Invited presentation.
- International Symposium on Mathematical Programming (ISMP), Montreal, Canada, 2024. Invited presentation.
- INFORMS Annual Meeting, Phoenix, AZ, 2023. Invited presentation.
- INFORMS Annual Meeting, Indianapolis, IN, 2022. Invited presentation.

HONORS AND AWARDS

Outstanding Doctor of Philosophy in Industrial Engineering Student, Texas A&M University, 2023.
Department Entry Scholarship, Texas A&M University, 2019.
Dean's Honors List, Faculty of Engineering, The Hong Kong Polytechnic University, 2013–2017.
Commercial Radio 50th Anniversary Scholarship, 2016.
Department Entry Scholarship, The Hong Kong Polytechnic University, 2013.

ACADEMIC SERVICE

Conference Session Chair

- Modeling for Mitigation: Designing Disease Control Protocols, INFORMS Annual Meeting, 2025.
- Strengthening Healthcare Systems for Preparedness, INFORMS Annual Meeting, 2024.
- Breaking the Chain: Using OR to Control Infectious Outbreaks, INFORMS Annual Meeting, 2023.

Journal Peer Reviewer

INFORMS Journal on Data Science; *Health Care Management Science*; *INFOR: Information Systems and Operational Research*; *Scientific Reports*; *AI and Ethics*; *Journal of King Saud University - Computer and Information Sciences*; *Discover Global Society.*

TEACHING EXPERIENCE

Instructor
Rice University, Houston, TX *Aug 2025–Dec 2026*
Courses: CMOR 462 Optimization in Finance (Fall 2026); CMOR 463 Operations Research in Healthcare

Teaching Assistant
Texas A&M University, College Station, TX *May 2022–Dec 2023*
Courses: ISEN 620/320 Survey of Optimization/Operations Research I; ISEN 340 Operations Research II; ISEN 302 Economic Analysis of Engineering Projects; ISEN 230 Informatics for Industrial Engineers